# Week 16

**Course Level Objectives:**

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

| Module Objectives | Instructional Materials | Learning Activities | Assessments | Tools |
|---|---|---|---|---|
| **1.** Select the appropriate procedure for a research question among one- and two-proportion z, one- and two-sample t, paired t, chi-square, ANOVA, and regression. (CLO 8)<br>**2.** Perform a complete hypothesis test, from hypotheses to a conclusion in context. (CLO 8)<br>**3.** Construct and interpret a confidence interval, using simulation or theoretical distributions as appropriate. (CLO 6)<br>**4.** Evaluate whether a study's design allows its conclusion to be generalized to a population. (CLO 1)<br>**5.** Evaluate whether a study's design supports a cause-and-effect conclusion. (CLO 1) | Review Slides: Putting It All Together (with answer key and no-answers versions)<br>All Slides and Interactive Lectures (Weeks 1–15)<br>StatKey | Participation: One Complete Test with a Partner – 10 problems, one per pair (narrator/writer roles), 5-minute presentations; extension: build the matching 95% CI | Final Exam – Wed Dec 9 (Blackboard, proctored, 2 hours, cumulative; no retake) (all objectives) | Blackboard<br>WileyPLUS<br>StatKey<br>Respondus |